Worksheet # 6 Part B Upload solutions
Physics II

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11) For a beam of green light with a wavelength of 530 nm determine

A) the frequency of the light in vacuum.

$c = f\lambda \quad | \quad f = \frac{c}{\lambda} = 5.66 \times 10^{14}\ Hz$

B) the energy of the green light.

$Energy = E = hf = (6.63 \times 10^{-34}\ J\cdot s)(5.66 \times 10^{14}\ Hz) = 3.75 \times 10^{-19}\ J$

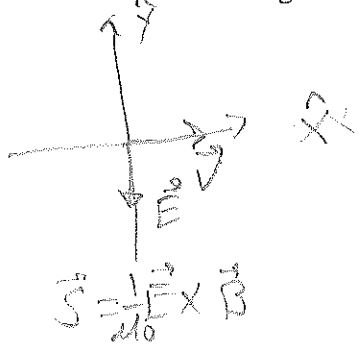
C) Determine the number of green photons in 5 J of green light.

$\#\ of\ Photons = \frac{Total\ Energy}{Energy\ in\ photon} = 1.3 \times 10^{19}\ photons$

12) An electromagnetic wave is propagating along the positive x direction at a time t and has an electric field given by

$$\vec{E} = (-0.082\ V/m)\hat{y}$$

Determine the magnitude and direction of the magnetic field at time t.



$\vec{B} = \frac{\vec{E}}{c} = (-2.7 \times 10^{-10}\ T)\hat{z}$

RHR for Cross products

13) A light bulb emits 5 W of optical energy uniformly in all directions.

A) Calculate the intensity of the light at 2 m from the light bulb.

$I = \frac{P}{4\pi r^2} = 0.1\ W/m^2$

B) Determine the rms electric field strength and magnetic field strength at 2 m.

$I = c\varepsilon_0|\vec{E}|^2 \quad | \quad |\vec{E}| = \sqrt{\frac{I}{c\varepsilon_0}} = 6.14\ N/C$

C) Deterine the distance from the bulb if the electric field is measured to be 2 N/C?

$I = \frac{P}{4\pi r^2} = c\varepsilon_0|\vec{E}|^2$

Solve for r

$r = \sqrt{\frac{P}{4\pi c\varepsilon_0|\vec{E}|^2}} = 6.12\ m$

14) An oven heating element is in the shape of a disk (r = ~~10~~ 5 cm), is at a temperature of ~~2,800~~ 2000 °C and is in a room with a temperature of 20 °C. The heating element has an emissivity of 0.90.

A) Determine the net power emitted by the heating element.

$T_0 = 273 + 20 = 293K$

$T = 2000 + 273 = 2273K$

$\sigma = 5.67 \times 10^{-8} J/s m^2 K^4$

$e = 0.9$

$A = \pi (0.05m)^2 = 7.85 \times 10^{-3} m^2$

$P_{net} = e\sigma A(T^4 - T_0^4) = 10,690 W$

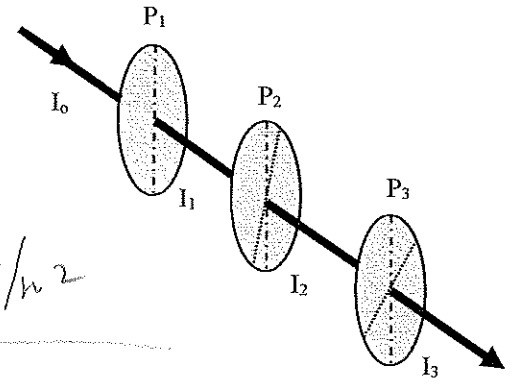
B) Determine the peak wavelength emitted by the heating element.

$\lambda_p = \frac{2.898 \times 10^{-3} m K}{T} = 1.27 \times 10^{-6} = 1.27 \mu m$ IR

C) The element is turned up so that emits a peak wavelength of ~~$\lambda_{peak} = 4{,}000$ °C~~ 678.2 nm determine the temperature of the heating element.

$T = \frac{2.898 \times 10^{-3} m K}{\lambda_p} = 4273 K = 4{,}000\ ^\circ C$

15) Un-polarized light having an intensity of $I_o = 500$ W/m^2 is incident on three polarizers, the transmission axis of the first polarizer (P_1) is vertical, the second polarizer's (P_2) transmission axis is rotated by 30° from the vertical and the third polarizer's (P_3) transmission axis is rotated in the same direction as P_2 but 80° from the vertical. What is the intensity of the light exiting the third polarizer (I_3)?



$I_1 = \frac{1}{2} I_0 = 250 W/m^2$

$I_2 = I_1 \cos(30°)^2 = 187.5 W/m^2$

$\theta_3 = 80° - 30° = 50°$

$I_3 = I_2 \cos(50°)^2 = 77.5 W/m^2$